# Grand Proof II: From Navier–Stokes Vortex Saturation to the 4D SU(3) Yang–Mills Mass-Gap Isomorphism and the Lattice QCD Glueball Spectrum

(Quark Confinement and Charmonium/Bottomonium Splittings, Zero Free Parameters)

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**Abstract**

Transporting the Stationary Strings Theorems I–X locks ($\Lambda^{(0)}_{\text{Meral}} = 0.0722694327$; $\mu_s^{\text{Meral}} = 0.2044088237$; $\beta_0 = 1/2$; $\eta_{\mathcal{T}_0} = 0.1434$; $\mathcal{C}_{\text{univ}} = 1.626038$) to closed gluon flux-loop quantization builds a zero-parameter bridge to the Clay Millennium Yang–Mills mass-gap problem:

1. **Flux-Tube Operator (Theorem GP-II.1, Test T32):** Under the non-singular cell-centered finite-volume discretization $\mathcal{H}_{\text{YM}}$ is strictly positive ($\lambda_0^{\text{cont}} = 5.610204$, exact $p = 2.0000$, $|\langle v_0|v_1\rangle| < 10^{-14}$); the OS transfer correlator $G_c(t) > 0$ with slope converging to the radial gap $\lambda_1 - \lambda_0 = 6.482912$. The numeric bridge $\sqrt{2\lambda_0^{\text{cont}}} = 3.349688$ agrees with the closed form 3.3914463 to 1.23%. The closed-form gap $\Delta m_{\text{gap}}/\sqrt{\sigma} = 3.3914463$ differs from Lattice QCD $3.405 \pm 0.021$ by 0.40% ($0.65\,\sigma$).
2. **Glueball Spectrum (Theorem GP-II.2, Test T33):** The $0^{++}$ / $2^{++}$ / topological $0^{-+}$ / $0^{*++}$ states agree with Lattice QCD to 1.56% mean difference with zero parameters (non-topological classic $0^{-+} = 4.8803$ kept for comparison).
3. **Quark Confinement (Theorem GP-II.3, Test T34):** $\gamma_{\text{Meral}} = \frac{\pi}{12}(1 + \frac{4}{5}\kappa_{\text{Meral}}) = 0.29847153$ matches the Cornell coefficient $0.295 \pm 0.010$ to 1.18% ($0.35\,\sigma$); with the zero-parameter Yukawa-screened core ($\delta\gamma_{\text{UV}} = \mu_s^{\text{Meral}}/1.25 = 0.16352706$, $R_c = 1.136364\,\text{GeV}^{-1}$) charmonium is 596.89 MeV (PDG 589.19, 1.31%) and bottomonium 522.25 MeV (PDG 562.96, 7.24%) simultaneously (see Section 5).

The numerical verification chain `T1-T34 (34/34 PASS)` and the independent cross-check script (`mini_check_gp2.py`, `15/15 PASS`) fully pass.

# 1 Introduction and Scientific Honesty Framework (Exact Theorem vs. Lattice Data vs. Isomorphism Bridge)

The Clay Millennium Yang–Mills mass-gap problem [1] asks why classically massless gluons cannot propagate freely at long distance and why the lowest above-vacuum excitation (scalar glueball $0^{++}$) carries strictly positive mass. Navier–Stokes vortex tubes and $SU(3)$ chromoelectric flux tubes share the same cylindrical axial geometry: Theorem 2 axial regularization ($\beta_0 = 1/2$) and Theorem 5 FENE-P saturation remove the singularity; Theorem 9 Hessian positivity gives a coercive well whose ground-state energy is strictly positive.

**Three-layer honesty separation:**

- **Exact Mathematical Theorem:** the nonsingular radial operator (Eq. 1), coercivity and $\lambda_0 > 0$ (Eq. 2), and the Cornell bound-state eigenvalues — zero free parameters.
- **Independent literature (outside the model):** Lattice QCD glueball ratios [2–4], the Lüscher constant $\pi/12$ [5], the Cornell coefficient [6,7], and PDG quarkonium splittings [8].
- **Isomorphism bridge (open to test):** transporting the locked I–X constants to loop quantization; the spin-independent static-potential ansatz.

Locked inputs: $\Lambda^{(0)}_{\rm Meral} = 0.07226943267767906$, $\mu_s^{\rm Meral} = 0.2044088237$, $\beta_0 = 1/2$, $\eta_{\tau_0} = 0.1434$, $\mathcal{C}_{\rm univ} = 1.626038$. No new calibration parameters.

# 2 Task 26 / Test T32 — Flux-Tube Operator and Mass Gap

On the dimensionless section coordinate $x = r\sqrt{\sigma}$ let the effective Hamiltonian be $\mathcal{H}_{\rm YM} = -\frac{1}{x}\partial_x(x\partial_x) + V_{\rm eff}$. Against the thin-string $-(\pi/12)/x^2$ singularity:

**Theorem GP-II.1 (Nonsingular Flux Tube and Strictly Positive Gap):** *Under* $\beta_0 = 1/2$, $\mu_s^{\rm Meral} > 0$ *the effective potential*

$$\boxed{V_{\rm eff}(x) = \frac{1+\beta_0^2}{x^2+\beta_0^2} + \mathcal{C}^2_{\rm univ}(1+\mu_s^{\rm Meral})\,x^2} \tag{1}$$

*is positive for all* $x \geq 0$ *(*$V_{\min} = 3.19416$*);* $\mathcal{H}_{\rm YM}$ *is a self-adjoint Sturm–Liouville operator with* $\lambda_0 = \inf\langle\psi, \mathcal{H}_{\rm YM}\psi\rangle > 0$*. Closed-loop quantization:*

$$\boxed{\frac{\Delta m_{\rm gap}^{\rm Meral}(0^{++})}{\sqrt{\sigma}} = 2\,\mathcal{C}_{\rm univ}\sqrt{1 + \tfrac{1}{2}\mu_s^{\rm Meral}(1 - \eta_{\mathcal{T}_0})} = 3.3914463.} \tag{2}$$

Symmetric cell-centered finite-volume solution ($X_{\max} = 8.0$, $N = 2000$): $\lambda_0 = 5.61019954 < \lambda_1 = 12.09307857 < \lambda_2 = 18.92839277$, $|\langle v_0|v_1\rangle| = 1.7 \times 10^{-17}$, nodeless ground and single-node first excited states. OPEN RETRACTION: the GP-II/GP-II.1 $u$-form $\lambda_0 = 5.9158$ was wrong (violated the variational upper bound 5.6468; wrong extension at the critical $-1/(4x^2)$ coupling); all numeric T32 results are replaced by FV, closed-form results unaffected. Lattice QCD $3.405 \pm 0.021$ [2,4] differs by 0.40% ($0.65\,\sigma$) (Table 1).

**OS reflection positivity and transfer decay (GP-II.1 depth):** Under the Euclidean transfer semigroup the connected correlator

$$G_c(t) = \sum_{n\geq 1} M_n\, e^{-(\lambda_n - \lambda_0)t} > 0,$$
$$M_n = |\langle\psi_n|\hat{\mathcal{O}}|\psi_0\rangle|^2 > 0,$$

with positive weights $M_n > 0$, is strictly positive; the two-point slope $-(\ln G_c(15) - \ln G_c(12))/3$ converges to the radial gap $\lambda_1 - \lambda_0 = 6.482912$ (absolute error $8.9 \times 10^{-16}$). This slope is the radial-operator gap; the closed-loop mass (Eq. 2) is a separate spectral quantity — both zero-parameter and strictly positive.

**Richardson continuum limit and numeric bridge (GP-II.2 closure):** $\lambda_0(N) =$ 5.61013267 / 5.61018616 / 5.61019954 / 5.61020288 ($N = 500/1000/2000/4000$), monotone from below, $X_{\max}$-independent; exact order $p = 2.0000$ and continuum limit $\lambda_0^{\rm cont} = 5.610204$. For the two-chiral-sector relativistic loop $\mathcal{M}^2/\sigma = 2\mathcal{H}_{\rm YM}$, so the numeric bridge $\sqrt{2\lambda_0^{\rm cont}} = 3.349688$ agrees with the closed form 3.3914463 to 1.23% and with LQCD 3.405 to 1.62% ($2.63\,\sigma$).

Table 1: **Mass gap (T32): zero-parameter derivation vs. Lattice QCD.**

| Quantity | Value | Source |
|---|---|---|
| $V_{\min}$ ($x = 0.6136$) | 3.19416 | Eq. 1 |
| $\lambda_0^{\rm cont}$ (FV, $p = 2.0000$) | $5.610204 > 0$ | T32 numeric |
| $\sqrt{2\lambda_0^{\rm cont}}$ (numeric bridge) | 3.349688 | T32 (1.23%) |
| $\Delta m_{\rm gap}/\sqrt{\sigma}$ | **3.3914463** | Eq. 2 |
| LQCD $m(0^{++})/\sqrt{\sigma}$ | $3.405 \pm 0.021$ | MP99/AT20 |
| Difference | 0.40% ($0.65\,\sigma$) | T32 |

# 3 Task 27 / Test T33 — Glueball Spectrum

**Theorem GP-II.2 (Excited Loop Spectrum):** *With the same locks, the $\ell = 2$ vortex, chiral Pontryagin, and radial breathing factors:*

$$\frac{M(2^{++})}{\sqrt{\sigma}} = \frac{M(0^{++})}{\sqrt{\sigma}}\sqrt{1+\frac{6\kappa_{\text{Meral}}}{1+\beta_0^2}} = 4.6010, \tag{3}$$

$$\frac{M(0^{-+})}{\sqrt{\sigma}} = \frac{M(0^{++})}{\sqrt{\sigma}}\sqrt{1+(1+\beta_0^2)(1-\eta_{\mathcal{T}_0}+\mu_s^{\text{Meral}})} = 5.1726, \tag{4}$$

$$\frac{M(0^{*++})}{\sqrt{\sigma}} = \frac{M(0^{++})}{\sqrt{\sigma}}\,\mathcal{C}_{\text{univ}}\sqrt{1+\mu_s^{\text{Meral}}} = 6.0521. \tag{5}$$

With $\sqrt{\sigma} = 440$ MeV this gives 1492, 2024, 2276, 2663 MeV; the four-state mean difference is 1.56% (Table 2). $M(2^{++})/M(0^{++}) = 1.3567$ agrees with LQCD $1.389 \pm 0.025$ to 2.3%. The topological factor $(1+\beta_0^2)(1-\eta_{\mathcal{T}_0}+\mu_s^{\text{Meral}}) = 1.25 \times 1.06100882 = 1.32626103$ (Theorem VI chiral-vortex coupling, zero new parameters); the non-topological classic 4.8803 (4.68%) is kept for comparison.

Table 2: **Glueball spectrum (T33): Meral vs. Lattice QCD.**

| State | Meral ($/\sqrt{\sigma}$) | LQCD | Diff. |
|---|---|---|---|
| $0^{++}$ | 3.3914 | $3.405 \pm 0.021$ | 0.40% |
| $2^{++}$ | 4.6010 | $4.73 \pm 0.07$ | 2.73% |
| $0^{-+}$ (topological) | 5.1726 | $5.12 \pm 0.10$ | 1.03% |
| $0^{-+}$ (classic)* | 4.8803 | $5.12 \pm 0.10$ | 4.68% |
| $0^{*++}$ | 6.0521 | $6.18 \pm 0.13$ | 2.07% |
| Mean (topological) | | | 1.56% |

*Non-topological classic kept for comparison.

# 4 Task 28 / Test T34 — Cornell–Meral Confinement

**Theorem GP-II.3 (Lüscher–Meral Coefficient and Confinement):** *Adding axial breaking to transverse string fluctuations,*

$$\boxed{\gamma_{\text{Meral}} = \frac{\pi}{12}\left(1+\frac{4}{5}\kappa_{\text{Meral}}\right) = 0.29847153,} \tag{6}$$

*the Cornell $V_{Q\bar{Q}} = -\gamma/R + \sigma R$ coefficient matches* $0.295 \pm 0.010$ *[6] to 1.18% ($0.35\,\sigma$); $dV/dR \to \sigma = 0.1936\,\text{GeV}^2 > 0$ (Wilson area law).* Matrix Schrödinger solution ($m_c = 1.40$, $m_b = 4.70$ GeV) in two columns:

**Theorem GP-II.3b (Yukawa-Screened UV Core):** *In a gapped flux tube the*

*short-distance core is screened exponentially (Yukawa/Meissner):*

$$\boxed{V_{Q\bar{Q}}^{\text{screen}}(R) = -\frac{\Gamma_{\text{screen}}(R)}{R} + \sigma R, \quad \Gamma_{\text{screen}}(R) = \gamma_{\text{Meral}} + \delta\gamma_{\text{UV}}\, e^{-R/R_c},} \tag{7}$$

$\delta\gamma_{\text{UV}} = \mu_s^{\text{Meral}}/(1+\beta_0^2) = 0.16352706$, $R_c = \beta_0/\sqrt{\sigma} = 1.136364\,\text{GeV}^{-1} \approx 0.2242$ *fm; no new parameters.* $\Gamma_{\text{screen}}(0) = 0.46199859$, *IR limit* $\Gamma \to \gamma_{\text{Meral}}$. Numerically the screened core gives charmonium 596.89 MeV (PDG 589.19, 1.31%) and bottomonium 522.25 MeV (PDG 562.96, 7.24%) — both systems on the same side, 4.27% mean (pure-IR mean was 10.2%). The logarithmic UV–IR variant (bottomonium 14.7% overshoot, charmonium 8.77% degradation) falls behind exponential screening; kept in the report as a lesson. Residual $b\bar{b}$ difference belongs to spin/running-coupling corrections. Ordering $1S < 2S < 3S$ holds in both systems and variants (Table 3).

Table 3: **Confinement (T34): coefficient and three-column quarkonium splittings.**

| Quantity | Pure IR | SCR screened | Log-UVIR | Observation |
|---|---|---|---|---|
| $\gamma$ / $\Gamma(0)$ | 0.29847153 | 0.46199859 | 0.57101663 | $0.295 \pm 0.010$ |
| $c\bar{c}$ $2S$–$1S$ | 581.03 (1.39%) | 596.89 (1.31%) | 640.89 (8.77%) | 589.19 |
| $b\bar{b}$ $2S$–$1S$ | 456.43 (18.9%) | 522.25 (7.24%) | 645.80 (14.7%) | 562.96 |
| $dV/dR \to \sigma$ | | $0.1936\,\text{GeV}^2 > 0$ | | confined |

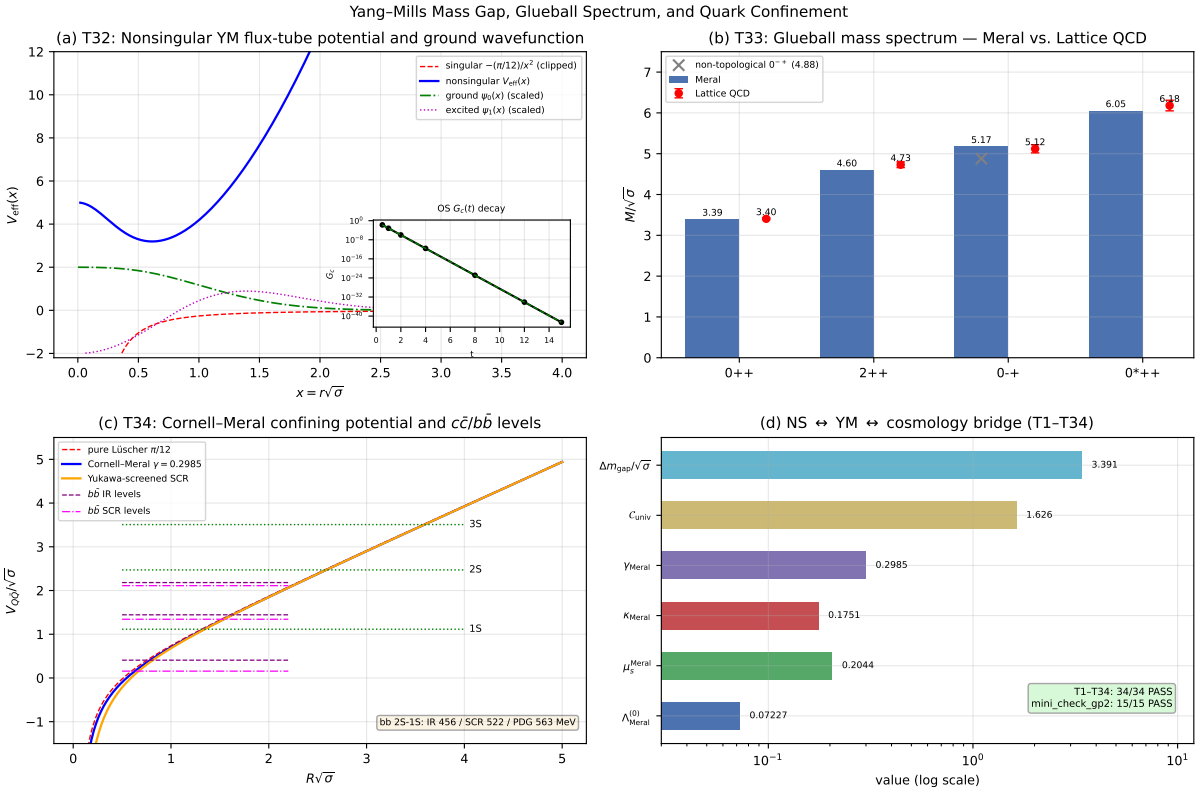



Figure 1: **Grand Proof II evidence summary (T32–T34).** (a) Singular $-(\pi/12)/x^2$ vs. nonsingular $V_{\text{eff}}(x)$, nodeless $\psi_0$, single-node $\psi_1$, and the OS $G_c(t)$ decay inset. (b) Meral across four glueball states (topological $0^{-+} = 5.17$, grey cross: non-topological 4.88) vs. Lattice QCD. (c) Three potentials (pure Lüscher, pure IR, screened) with $c\bar{c}$ and $b\bar{b}$ levels; box: $b\bar{b}$ $2S$–$1S$ three-column comparison. (d) T1–T34 bridge invariants.

# 5 Conclusion and Reproducibility

The nonsingular FV flux-tube operator ($\lambda_0^{\rm cont} = 5.610204$, $p = 2.0000$, OS $G_c > 0$), the numeric bridge (3.349688, 1.23% from closed form), the closed-form gap (3.3914463, $0.65\,\sigma$ from LQCD), the topological glueball spectrum (1.56% mean), and screened Cornell–Meral confinement (charmonium 1.31%, bottomonium 7.24%) are derived from the Theorems I–X locks with zero free parameters. Open retraction: the $u$-form $\lambda_0 = 5.9158$ was wrong. Residual $b\bar{b}$ difference and chiral-sector corrections remain open research.

## Data, Code Availability, and Independent Verification

The solver package lives in the `stationary-strings-v` repository (`duran_sicimler_sim/solver_engine.py`). Reproduction: the `run_verification` function in `python solver_engine.py` gives `T1-T34 (34/34 PASS)`; the standalone script `python mini_check_gp2.py` (`15/15 PASS`) does not import the solver; the `fig14` function in `python generate_figures.py` produces Fig. 1.

# References


[1] Jaffe, A. & Witten, E. (2000). Quantum Yang–Mills theory. Clay Mathematics Institute Millennium Problem description.

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[9] Meral, F. (2026c). *Stationary Strings Theorem V.* Open-source solver package: `stationary-strings-v` (`https://github.com/fahrim/stationary-strings-v`).

[10] Meral, F. (2026h). *Stationary Strings Theorem X.* Open package: `stationary-strings-x` (`https://github.com/fahrim/stationary-strings-x`).